Example 8.4.1

Reaction Engineering Basics, The Book

A strong acid, A (15 M), at 20 °C is going to be neutralized using a strong base, B (15 M NaOH), also at 20 °C, producing a salt, S, and water, W, equation (1). If the acid and the base were simply mixed together, they would react violently, releasing enough energy to raise their temperature to over 225 °C. Instead, a 10 L SBSTR will be used to neutralize 5.0 L of the acid. The acid will initially be present in the reactor, and the base will be added a rate of 0.25 L min^{-1} until 5 L of the base have been added. The reacting fluid will exchange heat with 2.5 L cooling water in a perfectly mixed shell. Chilled water at 20 °C will enter the shell at a rate of 2.5 L min^{-1}. The heat transfer area is 2150 cm^2 and the heat transfer coefficient is 73 cal cm^{-2} h^{-1} K^{-1}. At the time the acid is charged to the reactor, the cooling water temperature within the heat exchanger is 20 °C. The density and heat capacity of the acid solution, base solution, and cooling water can all be taken to be those of water, 1 g cm^{-3} and 1 cal g^{-1} K^{-1}. The neutralization reaction, equation (1), is irreversible with a heat of reaction equal to –13.7 kcal mol^{-1}. The reaction is first order in both acid and base with a pre-exponential factor of 8.11 x 10^{12} L mol^{-1} s^{-1} and an activation energy of 17.7 kcal mol^{-1}, equation (2). The pressure in the reactor will be constant and equal to 1 atm. Plot the acid concentration and the reactor temperature as functions of time while the base is being added.

$$A + B \rightarrow S + W \tag{1}$$

$$r_1 = k_1 C_A C_B \tag{2}$$